Optimization Methodologies: Approximation Algorithms, Heuristics, and Metaheuristics

[A Formal Comparative Synthesis of Computational Frameworks]

Source: Answers of the Presentation_5th.pdf & Supplementary Optimization Texts

Presentation Outline: Ω

$\{\omega_1$: Approximation Algorithms (Bounded Theoretical Concepts)

ω_2: Heuristics (Functional Methodologies & Local Search)

ω_3: Metaheuristics (High-Level Conceptual Frameworks)

ω_4: Comparative Analysis (Synthesis Matrix)

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Source: Answers of the Presentation_5th.pdf, Section Contents

Section 1

Approximation Algorithms

$$P \neq NP \Rightarrow t_{exact} = O(2^n)$$

- To achieve practical polynomial time $O(n^k)$, exact optimality must be relaxed to a rigorously bounded proximity.

Formal Definitions: Approximation Frameworks

Approximation Algorithm

- A polynomial-time algorithm for an NP-hard optimization problem that yields a feasible solution with a provable bound relative to the optimum.

Approximation Ratio (ρ)

- Minimization:
  $\rho \geq 1 \Rightarrow \rho = \max\limits_{\text{instances}} \frac{\text{Algorithm Cost}}{\text{Optimal Cost}}$
- Maximization:
  $\rho \leq 1 \Rightarrow \rho = \min\limits_{\text{instances}} \frac{\text{Algorithm Profit}}{\text{Optimal Profit}}$

PTAS (Polynomial-Time Approximation Scheme)

- Family of algorithms parameterized by $\epsilon > 0$. Returns a solution within $(1+\epsilon)$ of optimum. Runs in polynomial time for fixed ϵ (e.g., $O(n^{1/\epsilon})$).

FPTAS (Fully PTAS)

- Stronger formulation. Running time is polynomial in both the input size n and $1/\epsilon$ (e.g., $O(n^2/\epsilon)$).

Numerical Example: Vertex Cover Approximation

Problem Formulation:
Graph $G = (V, E)$.
Objective: $\min |C|$ subject to $\forall (u, v) \in E, u \in C \vee v \in C$.
Initial State $E' = \{ (1, 2), (1, 3), (2, 3), (2, 4), (3, 5), (4, 5)\}$.

Evaluation State (Greedy Matching):
Select disjoint edges to form Matching $M = \{(1, 2), (3, 5)\}$.
Add both endpoints to Cover C.

Result & Ratio Proof:
Algorithm Output: $C = \{1, 2, 3, 5\} \Rightarrow |C| = 4$.
Known Optimum: $OPT = \{2, 3\} \Rightarrow |OPT| = 2$.
Approximation Ratio Verified: $\frac{|C|}{|OPT|} = \frac{4}{2} = 2$ (Matches theoretical bound $\rho \leq 2$).

Section 2

Heuristics

Source: Answers of the Presentation_5th.pdf, Section 2

$$t_{heuristic} \ll t_{exact} \Rightarrow \text{Guarantee} = \emptyset$$

- ❑ Sacrificing provable optimality guarantees to secure rapid, scalable, and computationally feasible solutions in unbounded search spaces.

Source: Answers of the Presentation_5th.pdf, Conceptual Questions 2.1

Formal Definitions: Heuristic Classifications

Constructive Heuristics

Definition: Builds a feasible solution strictly incrementally from an empty initial state.

Formulation: $S_0 = \emptyset \rightarrow S_{k+1} = S_k \cup \{e_i\}$, stopping when S_{final} is completely feasible.

Search-Based Heuristics

Definition: Iteratively modifies an existing complete feasible solution to locate a local optimum.

Neighborhood Structure $N(S)$: A defined set of nearby solutions reachable via a single local transformation operator from current state S.

Local Optimum: A state S^* where $\forall S' \in N(S^*), f(S^*) \leq f(S')$ (for minimization).

Source: Answers of the Presentation_5th.pdf, Conceptual Questions 2.1 & 2.2

Numerical Example: Constructive Heuristic (Knapsack)

Problem Statement:

Objective: $\max \sum_i v_i x_i$ subject to $\sum_i w_i x_i \leq W$.

Constraint: Capacity $W = 50$.

Component State (Greedy Ratio Evaluation):

Items Vector $I = [i_1, \quad i_2, \quad i_3]$,

Value Vector $V = [60, \quad 100, \quad 120]$,

Weight Vector $W_{item} = [10, \quad 20, \quad 30]$,

Score Ratio $(V/W) = [6, \quad 5, \quad 4]$.

Final Constructive State:

Selection Sequence: $S_0 = \emptyset \to S_1 = \{i_1\} \to S_2 = \{i_1, i_2\}$.

Result: Knapsack load = 30 $\leq$ 50. Total Value = 160.

Source: Answers of the Presentation_5th.pdf, Section 2.2

Section 3

Metaheuristics

$$P(\text{accept worse state}) > 0 \Rightarrow \text{Escape } N(S^*)_{local}$$

- Introducing stochastic perturbations and high-level control strategies to continuously balance vast global exploration with acute local exploitation.

Source: Answers of the Presentation_5th.pdf, Conceptual Questions 3.1

Formal Definitions: Metaheuristic Frameworks

General Framework

Definition: A high-level strategy governing subsidiary heuristics to escape local optima via diversification and intensification.

Simulated Annealing (SA)

Mechanics: Mimics physical thermodynamics. Probabilistically accepts worsening moves to escape local minima.

Acceptance Probability Axiom: $P(accept) = e^{-\Delta f/T}$, where T is the continuously decreasing system temperature.

Genetic Algorithms (GA)

Mechanics: Maintains a population matrix of candidate solutions, applying biologically-inspired transformations.

Operators: Selection (fitness-proportionate), Crossover (recombination), Mutation (stochastic variance).

Source: Answers of the Presentation_5th.pdf, Section 3.2

Numerical Example: GA Selection Components

Objective: Map a population's discrete fitness evaluations to selection probabilities.

Function: $P(x_i) = F(x_i) / \sum_{j=1}^{n} F(x_j)$.

Evaluation States (Vectors):

Population Vector
$S = [s_1, s_2, s_3, s_4, s_5, s_6]$

Fitness Vector $F = [12, 4, 16, 8, 36, 24]$

Total Fitness Sum: $\sum F = 100$

Probability Mapping Result (Aligned Vectors):

Selection Probability Vector
$P_{sel} = [0.12, 0.04, 0.16, 0.08, 0.36, 0.24]$

Proof:

Highest fitness s_5 (36) secures the highest geometric probability bounding (36% selection chance).

Section 4

Comparative Analysis

Source: Answers of the Presentation_5th.pdf, Section 4

$$F(\text{Quality, Time, Scalability, Guarantee}) \rightarrow \text{Optimal Strategy}$$

- **Methodological selection requires a rigorous synthesis** of constraint geometries, **computational** budgets, and tolerance for sub-optimality.

Diagnostic Matrix: Methodological Methodological Trade-Offs

Method	Solution Quality	Time	Scalability	Theoretical Guarantee
Exact Algorithms	Optimal	Exponential (NP-Hard)	Weak	Strongest
Approximation Algorithms	Near-optimal with bound	Polynomial	Good	Provable Ratio (ρ)
Heuristics	Good in practice	Usually very fast	Very good	Usually none
Metaheuristics	Often very good in practice	Moderate to high	Very good	Usually none or weak

Source: Answers of the Presentation_5th.pdf, Section 4.1

Methodological Synthesis

- **Axiom 1 (Theoretical Bounds):** Approximation algorithms uniquely bridge the $P \neq NP$ to queat divide by providing mathematically provable bounds (ρ) on solution quality in polynomial time.
- **Axiom 2 (Operational Velocity):** Simple Heuristics completely abandon theoretical guarantees to prioritize computational speed, utilizing direct, problem-specific logic for immediate feasibility.
- **Axiom 3 (Global Robustness):** Metaheuristics serve as overarching, stochastic control mechanisms designed to violently disrupt local optima, adapting to massive, complex search spaces where exact methods critically fail.

Source: Answers of the Presentation_5th.pdf, Conclusion